



Politecnico
di Bari



DIPARTIMENTO DI
INGEGNERIA ELETTRICA
E DELL'INFORMAZIONE



Assessing the impact of multimodal-by-design embeddings in recommender systems

Master Degree in Computer Engineering (AI and Data Science)

Dissertation in Foundations of Machine Learning

Supervisor

Prof. Tommaso Di Noia

Co-supervisors

Dr. Matteo Attimonelli

Dr. Claudio Pomo

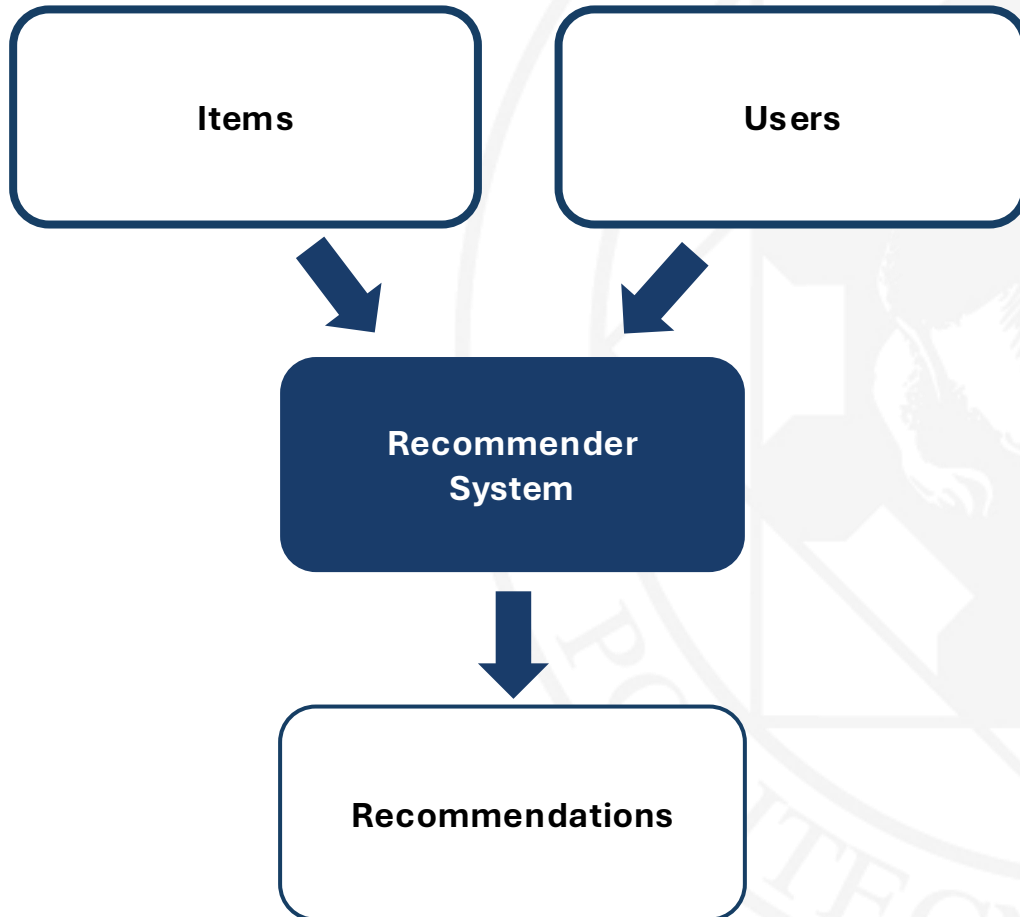
Candidate

Simone Bonfrate

Summary

- Introduction to problem
- Thesis Work
- Results
- Conclusions

Introduction – Recommender Systems



A **Recommender System** is an information filtering framework that provides personalized item recommendations based on user preferences (e.g., Amazon, Netflix, Facebook).

Main paradigms:

1. **Content Based Filtering:** recommendations based on item-level feature similarity
2. **Collaborative Filtering:** recommendations based on user–user or item–item interaction patterns

Introduction – Collaborative Filtering Issues

Collaborative Filtering Limitations:

- Data sparsity: limited user–item interactions
- Cold start: new users/items lack historical data

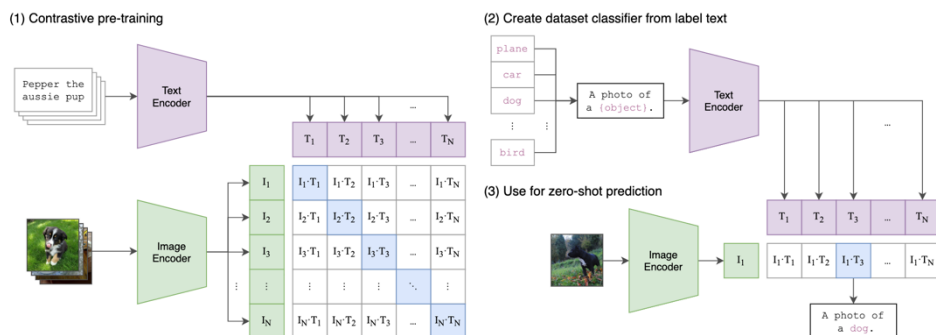
Solution: Hybrid approaches; integrate auxiliary signals :

- Side Informations (text, images, metadata)
- High Order relational signals

Multimodal Recommender System (MMRS)

- leverage multimodal item data to enhance representation quality
- mitigate sparsity and cold-start issues

Introduction – Visual Language Models as Feature extractor



MMRS rely on unimodal feature extractors (ResNet, SBERT):

- Requiring fusion mechanisms (Early/Late/Hybrid)
- May introduce information loss and weak cross-modal alignment

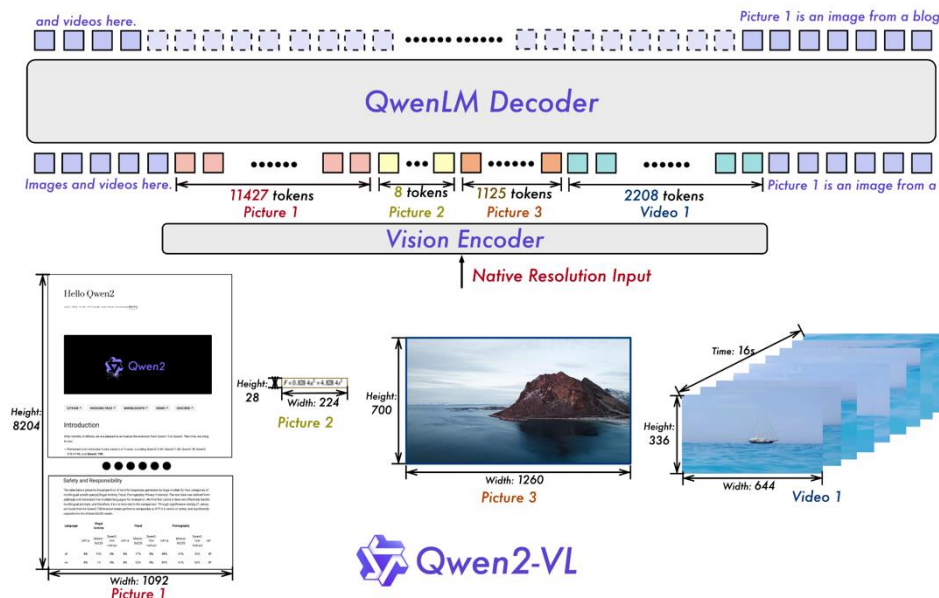
Visual Language Models (VLMs)

- Learn shared latent spaces across modalities
- Align textual and visual representations
- Improve semantic consistency

Large Vision-Language Models (LVLMs)

- Extend LLMs with visual encoders
- Learn unified multimodal representations

Examples: Qwen-VL, Phi-Vision



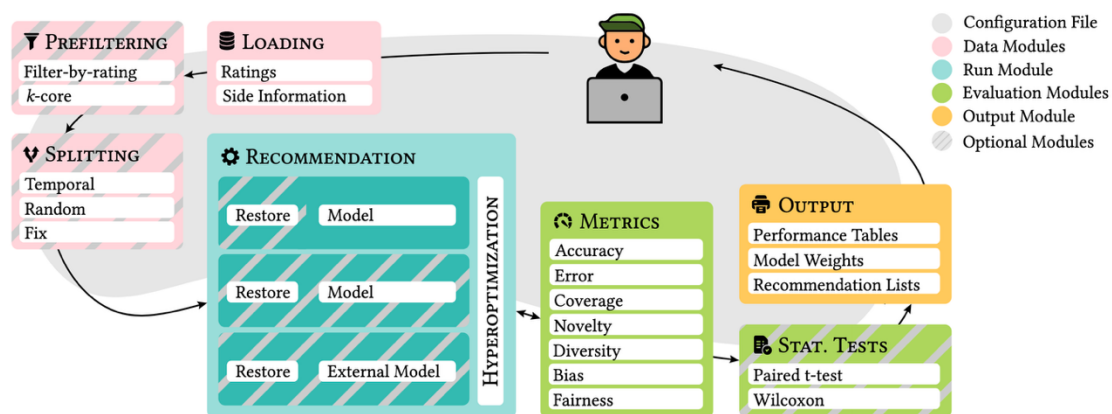
Thesis Objective

Investigate whether Large Vision-Language Models
can serve as effective and generalizable feature extractors
for Multimodal Recommender Systems (MMRS)

Thesis Work – Research Questions

- RQ1: Do the benefits of LVLM-based item representations generalize across domains and recommendation architectures?
- RQ2: Do newer LVLMs provide better item representations for recommendation than earlier LVLMs?
- RQ3: How does recommendation performance vary with the size and capacity of LVLMs?
- RQ4: Are generalist multimodal embedding models a better alternative to LVLMs for item representation in recommendation?
- RQ5: To what extent is recommendation performance affected by the prompting strategy used to extract LVLM representations?

Thesis Work – Metodology



Feature Extractor: ResNet50-SentenceBERT, ResNet50 only, ViT, CLIP, Qwen-VL, Phi, Qwen-VL-Embeddings, MM-Embed, GME.

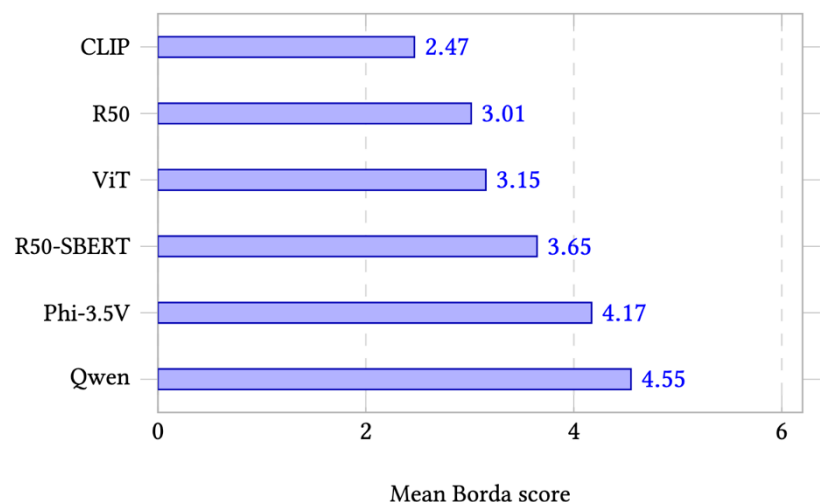
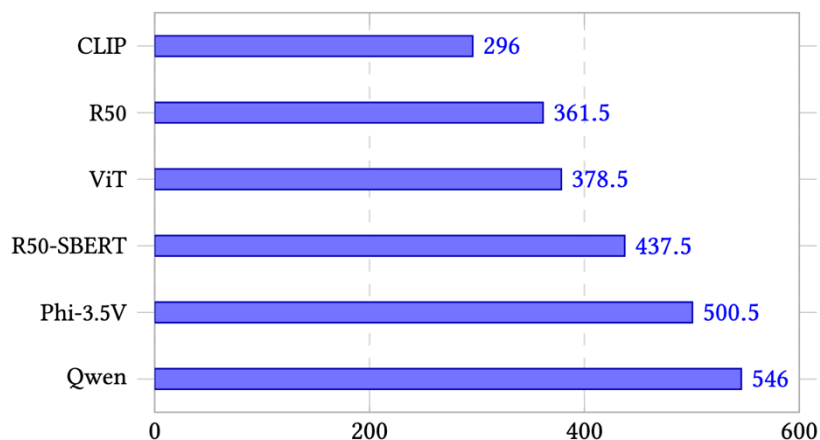
Feature Extraction Environment: Ducho framework and custom LVLM-GME extraction framework

Datasets: Amazon (Baby, Pets, Clothing), Movielens, Allrecipes

MMRS: VBPR, LATTICE, FREEDOM, BM3, LGMRec, COHESION, PGL, SMORE.

Recommendation Framework: Elliot (VBPR, LATTICE, FREEDOM, BM3) and MMRec (LGMRec, COHESION, PGL, SMORE)

Results – Do the benefits of LVLM-based item representations generalize across domains and recommendation architectures?



LVLM-based representations

- consistently rank among top-performing methods
- robust across datasets and architectures

Traditional extractors remain competitive

- ResNet50-SBERT, ResNet50, ViT outperform LVLMs in specific settings

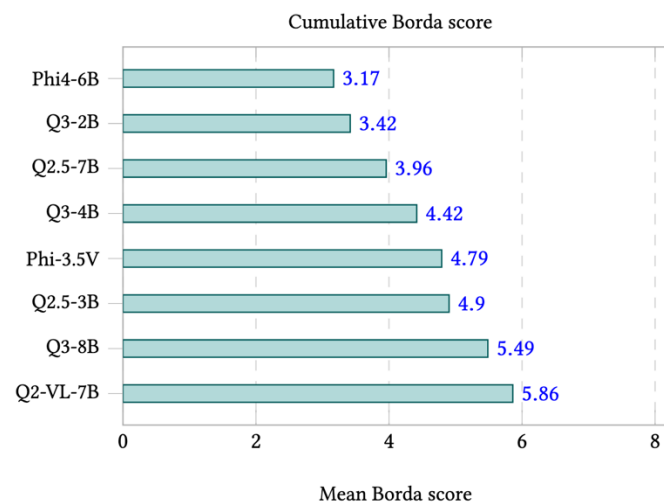
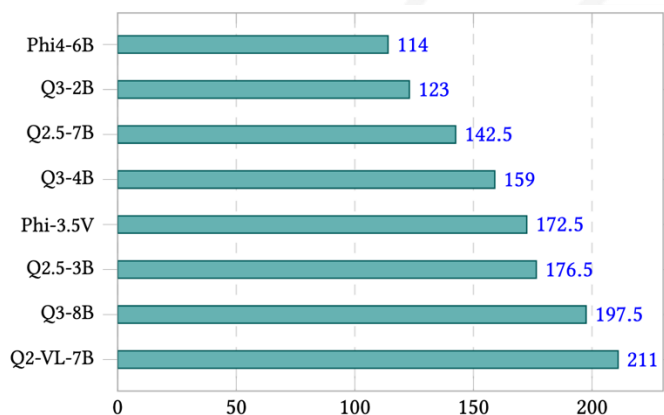
Simple VLMs (e.g., CLIP) underperform

- cross-modal alignment alone is insufficient

Key Insight:

- Performance depends on interaction between representation and model inductive bias

Results – Do newer LVLMs provide better item representations for recommendation than earlier LVLMs?



Earlier LVLMs already encode strong semantic representations

- Well aligned with recommendation tasks

Newer LVLMs do not consistently outperform

- No monotonic improvement trend

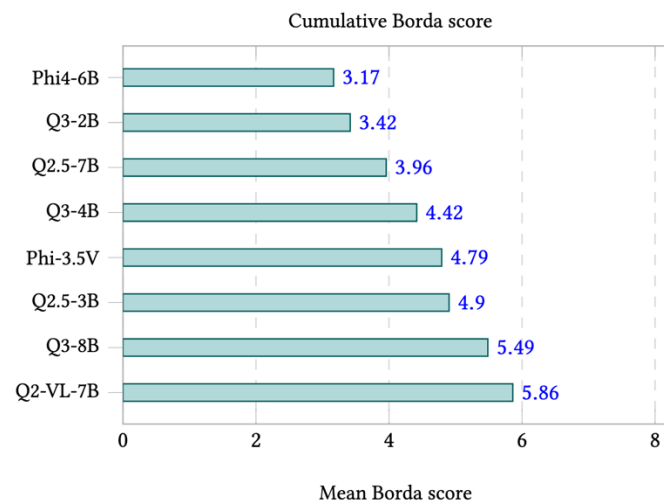
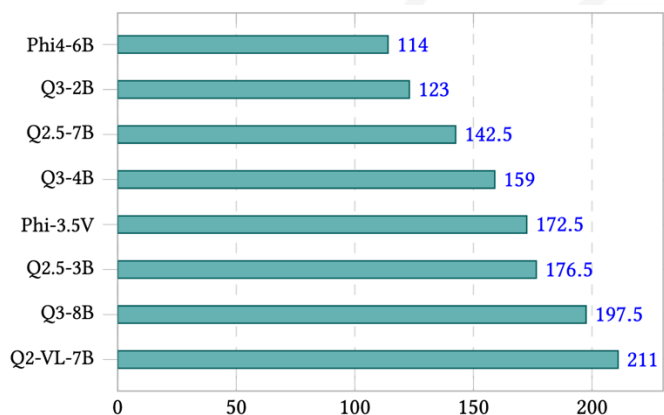
Gains are configuration-specific

- Not systematically reproducible

Key Insight:

- Model scaling and architectural evolution do not guarantee improved recommendation effectiveness

Results – How does recommendation performance vary with the size and capacity of LVLMs?



No monotonic scaling behavior

- Performance does not consistently improve with size

Increasing parameters $\neq$ better performance

- Larger models produce more expressive but harder-to-optimize representations

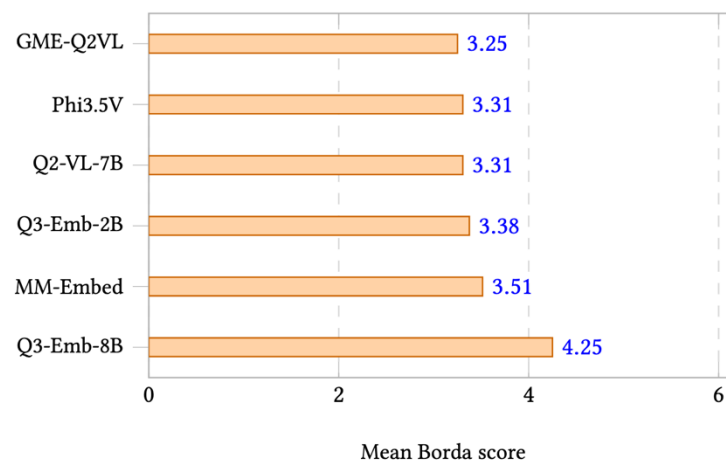
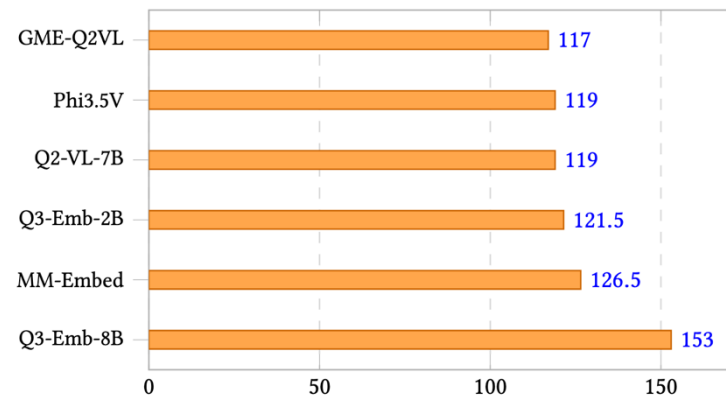
Smaller models often outperform larger ones

- Capacity alone is not a reliable indicator of effectiveness

Key Insight

- There exists an optimal capacity regime for recommendation

Results – Are generalist multimodal embedding models a better alternative to LVLMs for item representation in recommendation?



GME models often achieve higher performance

- Particularly in similarity-based architectures

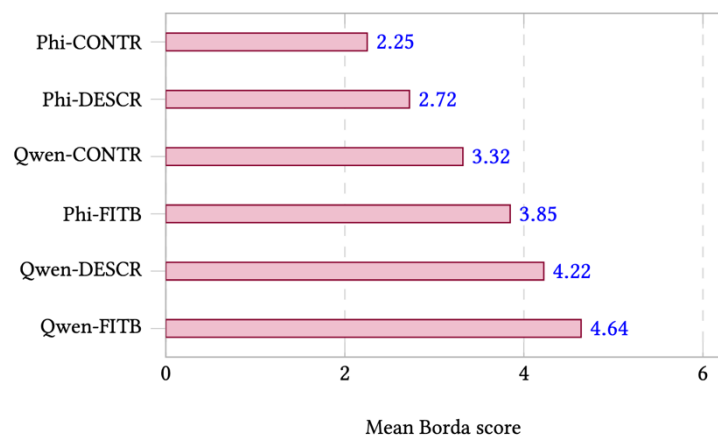
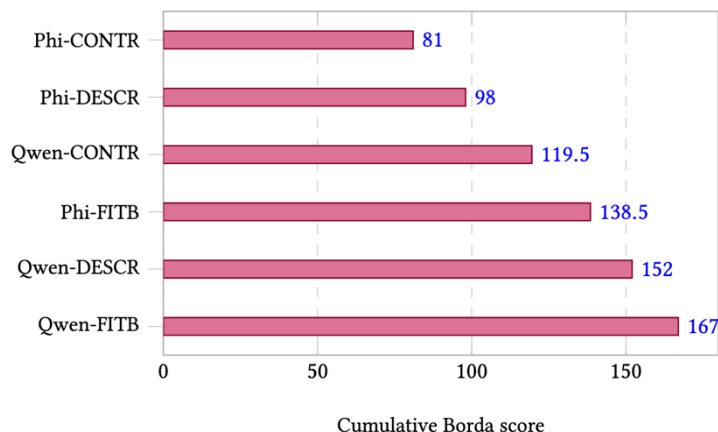
No consistent dominance

- LVLMs remain competitive in several configurations
- Performance depends on representation–model alignment

Key Insight:

- Effectiveness is driven by alignment with the recommender, not by the representation paradigm (LVLM vs GME)

Results – To what extent is recommendation performance affected by the prompting strategy used to extract LVLM representations?



FIB (Fill-in-the-Blank)

- Robust and consistently strong baseline

DESC and CONSTR outperform in specific settings

- Performance is highly sensitive to prompt design

LVLMs handle misaligned prompts

- Robust semantic realignment (e.g., Qwen2-CONTR)

Key Insight:

- Prompting shapes both representation quality and embedding space geometry

Conclusions

LVLMS provide semantically rich representations

- Strong and often state-of-the-art performance (but not universal)

GME models are highly competitive

- Outperform LVLMS in specific scenarios

Performance is driven by:

- representation learning
- model architecture
- prompting strategy

Future Work

- Develop recommendation-aware multimodal encoders
- End-to-end training with recommendation signals



Thanks for the attention